

Reproducible Manuscripts and Reports

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What is a Reproducible Report?

A **reproducible report** is a document that integrates data analysis, visualizations, and narrative text into a single cohesive output, which can be automatically regenerated and verified.

Key Components

- ▶ **Data:** Clearly sourced, documented, and accessible.
- ▶ **Code:** Transparent and executable by anyone.
- ▶ **Narrative:** Explanation and interpretation clearly linked to outputs.

Why Use Reproducible Reports?

- ▶ **Transparency:** Enhances trust and facilitates peer review.
- ▶ **Efficiency:** Easy updates when data or methods change.
- ▶ **Collaboration:** Simplifies teamwork and project continuity.

Tools for Creating Reproducible Reports

Open source:

- ▶ **Quarto**: Integrates R, Python, Julia and other languages directly into documents.
- ▶ **Jupyter Notebooks**: Supports Python and other languages for interactive analysis.

Output types

- ▶ Articles, presentations, dashboards, websites, blogs, and books in HTML, PDF, **MS Word**, **MS Powerpoint**, ePub, and more

Pdf

```
title: "My Document"
author: "Nicolas Reigl"
format:
  pdf:
    toc: true
    number-sections: true
    colorlinks: true
```

Beamer

Authoring (with Quarto)

Syntax is similar in Jupyter notebooks

Project: `_quarto.yml`

Listing 1 `_quarto.yml`

```
project:
  type: manuscript

manuscript:
  article: index.qmd

format:
  html:
    comments:
      hypothesis: true
  docx: default

# (other formats)
# pdf: default
```